THE ORIGIN OF WI-FI



By Mo Ike (They/Them)
and Yuanyuan Cui (She/Her)

Who Invented Wi-Fi?



Who Invented Wi-Fi?

John
O'Sullivan(?)



John O'Sullivan portrait by Steve Torchinsky. 2013. Wikimedia Commons

“Mother of Wi-Fi”



Hedy Lamarr in The Heavenly Body. Jewish Women's Archive

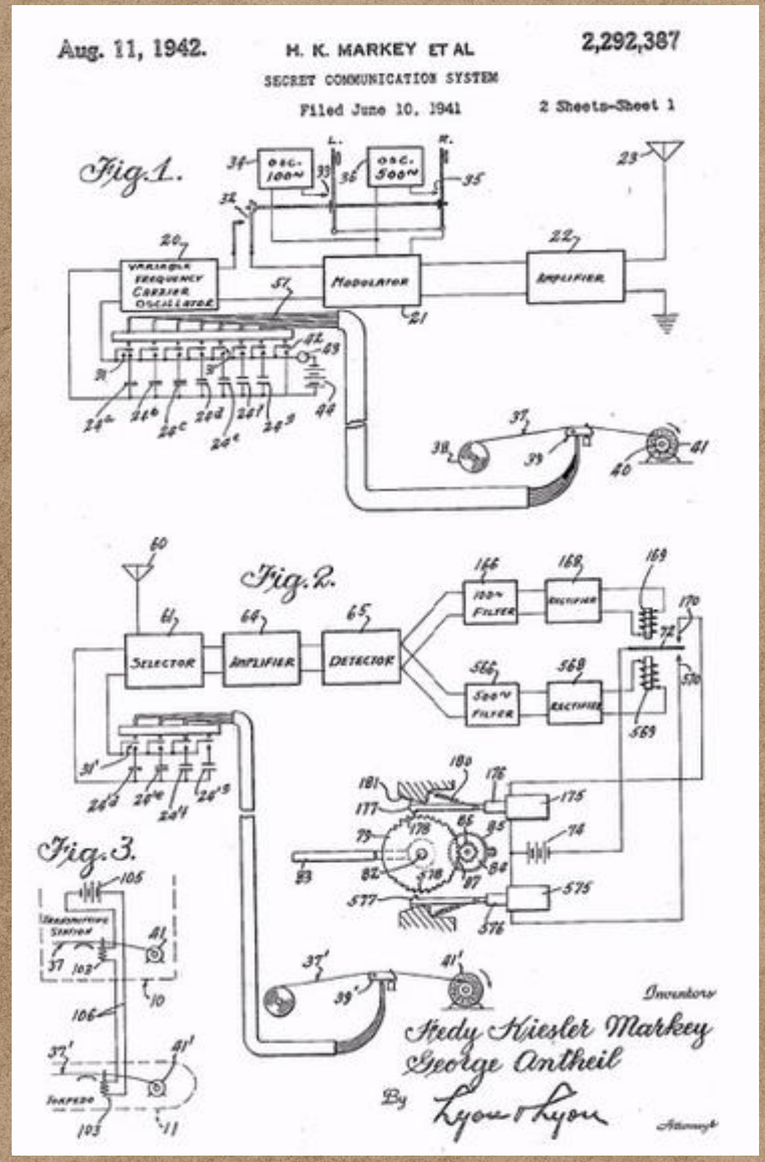
The avant-garde Composer



George Antheil portrait by Berenice Abbott. 1927. A&M Penn Photography Foundation. Wikimedia Commons

Secret Communication System (1941-1942)

- Early version of frequency hopping
- 88 frequencies (like a piano roll)
- US patent



Patent # 2,292,387 for a "Secret Communication System," granted to actress Hedy Kiesler Markey. National Air and Space Museum. National Archives

ALOHAnet (1971)

- Developed by a University of Hawai'i research team
- Led by Norman Abramson and Franklin Kuo
- Goal: connecting users on different campuses
- Random Access Protocol
- Inspired Carrier-Sense Multiple Access with Collision Avoidance (CSMA/CA)



User operating the ALOHAnet hub machine. University of Hawai'i at Mānoa. IEEE Spectrum

MICHAEL MARCUS

The rule Changer



Michael Marcus portrait.
[IEEE Microwave Theory and Technology Society]

The Old Rule

One frequency, one owner. Users needed an FCC license, and approval was slow and costly.

The Key: spread spectrum

Normal radio: one narrow sl[redacted]

Spread spectrum: thin and [redacted]

signal more resistant to noise and allows different users to share the same airwaves.

May 9, 1985

The Breakthrough
The FCC authorizes unlicensed spread-spectrum operation under technical rules.

900 MHz · 2.4 GHz · 5.8 GHz
No license needed. Main condition: use spread spectrum.

1997
First Wi-Fi standard was born at 2.4 GHz band

EARLY 1990S

The Need for Speed

3-6 Mibts/s
Wireless networks back then

100 Mibts/s
The goal: as fast as wired

The problem: echos

1. Signals bounce off walls, funiture and people
2. The same signal arrives many times, a tiny bit apart

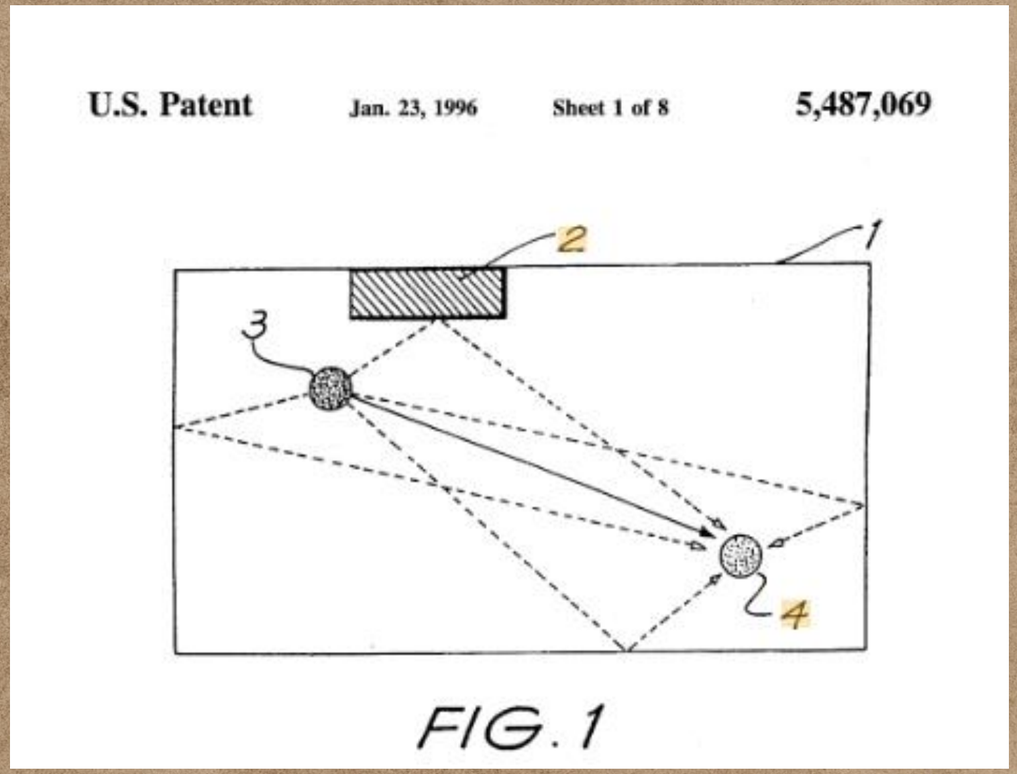


Fig. 1 from the “Wireless LAN” patent by John O’Sullivan et al. 1996. CSIRO. Google Patents

From Black Holes to Wi-Fi

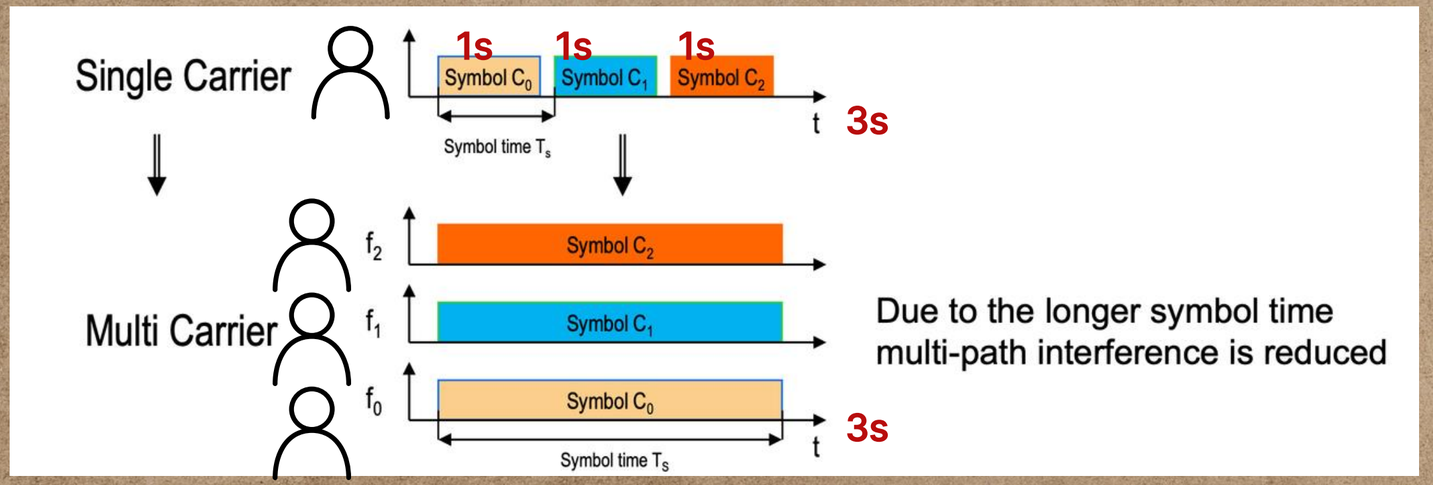


John O'Sullivan portrait by Steve Torchinsky. 2013. Wikimedia Commons

Radio astronomer's tool

The fast Fourier transform splits one signal into many frequencies, like a prism splits light.





Single carrier vs. multi carrier by Rick Daniel. RF Fundamentals Seminar. Rohde & Schwarz

Not New:

- FFT: an exsiting math tool
- Wireless network: they "already exist"

What his team did:

- Combined many channels with error correction
- Fit it all on affordable chips
- Patented the combination in 1992

1999

The Name Giver -Interbrands

The technical name

~~IEEE 802.11b Direct Sequence~~

Interbrand's pick

Wi-Fi

"Wi-Fi doesn't stand for anything."

Phil Belanger
Wi-Fi Alliance founding member. The name came from a list of 10 that Interbrand proposed.

CONCLUSION

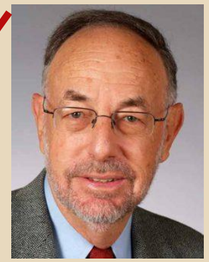
Who invented Wi-Fi ?





1942
The idea
Lamarr & Antheil

THE CASE
Wi-Fi



1985
The Airwaves
Michael Marcus



1971
The rules
ALOHAnet



1992
The Speed
John O'Sullivan

1999
The Name
Interbrand

Case closed or ...is it?

Bibliography (Part 1)

"ALOHAnet." Electrical & Computer Engineering University of Hawai'i, 2020, ee.hawaii.edu/home/page/ALOHAnet. Accessed 7 Oct. 2026.

CSIRO. "CSIRO Honours Wireless Team." CSIROpedia, 14 Oct. 2009, csiropedia.csiro.au/csiro-honours-wireless-team. Accessed 14 Oct. 2026.

Daniel, Rick. "Part 6: Intro to OFDMA (Orthogonal Frequency Division Multiple Access)." RF Fundamentals Seminar, Rohde & Schwarz, cdn.rohde-schwarz.com/ymkt/na/content/RF_fundamentals_seminar_materials/6_RF_Fun_-_Introduction_to_OFDMA.pdf. Accessed 4 Oct. 2026.

Doctorow, Cory. "WiFi Isn't Short for 'Wireless Fidelity.'" Boing Boing, 8 Nov. 2005, boingboing.net/2005/11/08/wifi-isnt-short-for.html. Accessed 5 Oct. 2026.

EPB. "The History of Wi-Fi: Who Invented It and When?" Epb.Com, 4 June 2025, epb.com/get-connected/tech-support/history-of-wifi-invention/. Accessed 19 Sept. 2026.

Erica Lansberg, DPhil. "Hedy Lamarr's WWII Invention Helped Shape Modern Tech." The National WWII Museum | New Orleans, The National World War II Museum, 23 Apr. 2025, www.nationalww2museum.org/war/articles/hedy-lamarrs-wwii-invention-helped-shape-modern-tech. Accessed 19 Sept. 2026.

Goodrich, Joanna. "ALOHAnet Introduced Random Access Protocols to the Computing World - IEEE Spectrum." IEEE Spectrum, 12 Aug. 2020, spectrum.ieee.org/alohanet-introduced-random-access-protocols-to-the-computing-world. Accessed 4 Oct. 2026.

Kabir, Syed Rafid. "Who Invented WiFi? Internet in the Air | History Cooperative." History Cooperative, 13 Dec. 2023, historycooperative.org/who-invented-wifi/. Accessed 14 Sept. 2026.

Markey, Hedy Kiesler, and George Antheil. Secret Communication System. nos. US-2292387-A, 11 Aug. 1942, patents.google.com/patent/US2292387. Accessed 4 Oct. 2026.

Bibliography (Part 2)

Marcus, Michael J. “Commercial Spread Spectrum Background.” Unlicensed Spread Spectrum Retrospective, IEEE 802 Plenary Session, 16 Mar. 2004, Walt Disney World, FL. Presentation slides, www.ieee802.org/802_tutorials/04-March/Revised-802-SS-talk.pdf. Accessed 27 Sep. 2026.

---. “Early Civil Spread Spectrum History.” Marcus Spectrum Solutions, www.marcus-spectrum.com/page4/SSHist.html. Accessed 27 Spt. 2026.

O’Sullivan, John D., et al. Wireless LAN. US Patent 5,487,069, 23 Jan. 1996. Google Patents, patents.google.com/patent/US5487069A/en. Accessed 4 Oct. 2026.

Photograph of Michael Marcus. IEEE Microwave Theory and Technology Society, mtt.org/profile/michael-markus/. Accessed 12 Oct. 2026.

Torchinsky, Steve. John O’Sullivan, near Hoogeveen, The Netherlands on the Occasion of the Colloquium “Reinventing Radio Astronomy” in Honour of Arnold van Ardenne. 29 May 2013, commons.wikimedia.org/wiki/File:John-OSullivan_20130528.jpg. Accessed 11 Oct. 2026, Photograph.

Technopolis Group. Presenting the (Economic) Value of Patents Nominated for the European Inventor Award 2012: Inventor File John O’Sullivan. European Patent Office, 30 Mar. 2012, link.epo.org/web/osullivan_en.pdf. Accessed 4 Oct. 2026.

Federal Communications Commission. Additional Spectrum for Unlicensed Devices Below 900 MHz and in the 3 GHz Band. Notice of Inquiry, ET Docket No. 02-380, FCC 02-328, 20 Dec. 2002, docs.fcc.gov/public/attachments/fcc-02-328a1.pdf. Accessed 27 Spt. 2026.

THANK YOU VERY MUCH

